

RVCalPy radial-velocity benchmark report

Broadening function: LL Aqr and AK For · Cross-correlation: 16 UVES high-velocity stars
RVCalPy, <https://github.com/atemelci/RVCalPy> · 2026 September 22

RVCalPy velocities are compared with published velocities of the same nights (LL Aqr, AK For) or the same spectra (UVES sample). **Broadening function:** LL Aqr scatter 0.092 / 0.142 km s⁻¹ (primary / secondary), χ_r^2 1.03 / 0.81; AK For scatter 0.150 / 0.864 km s⁻¹, χ_r^2 0.45 / 2.54, slope 0.9986 / 0.9923. **Cross-correlation:** 16 stars, median $|\Delta|$ 0.486 km s⁻¹, weighted offset $+0.207 \pm 0.125$ km s⁻¹, slope 1.000171 ± 0.000479 over 919 km s⁻¹.

1. What was tested

	LL Aqr	AK For	UVES sample
Method	Broadening function	Broadening function	Cross-correlation
Reference	Graczyk et al. (2016), Table 1	Helminiak et al. (2014), Table 2	Alazzawi et al. (2026), Table 3, V_r^{meas}
Instrument	HARPS	FEROS (14) + HARPS (2)	UVES, DIC1 390+580, red arm
Measured / compared	17 / 16 nights	16 / 14 nights	16 / 16 stars, one epoch each
Template	HIP 57757 (6093 K)	HIP 108870 (4676 K)	synthetic, 5215 K, log g 3.55
Window / step	500–550 nm, $\Delta v = 1$ km s ⁻¹	500–550 nm, $\Delta v = 2$ km s ⁻¹	500–550 nm
Period / velocity span	P = 20.1784 d	P = 3.98099 d	919 km s ⁻¹

Table 1. Material and settings. All spectra are public ESO archive phase-3 products; only epochs present in both data sets are compared.

2. How the numbers are defined

For each matched epoch the difference between our velocity and the published one is formed, with the two uncertainties added in quadrature. Δ is the weighted mean offset, s the scatter, χ_r^2 the reduced chi-square about Δ , and a , b the slope and intercept of the weighted straight-line fit; $a - 1$ is quoted as the amplitude (BF) or velocity-scale (CCF) error:

$$d_i = v_i^{\text{ours}} - v_i^{\text{pub}}, \quad \sigma_i = \sqrt{\varepsilon_i^{2, \text{ours}} + \varepsilon_i^{2, \text{pub}}}, \quad w_i = \sigma_i^{-2}$$

$$\Delta = \frac{\sum w_i d_i}{\sum w_i}, \quad s = \sqrt{\frac{\sum (d_i - \bar{d})^2}{N - 1}}, \quad \chi_r^2 = \frac{1}{N - 1} \sum \left(\frac{d_i - \Delta}{\sigma_i} \right)^2, \quad v^{\text{ours}} = a v^{\text{pub}} + b$$

No χ_r^2 is given for the UVES comparison: both velocities are measured from the same exposures and no per-star error is published (the paper quotes a global ~ 0.5 km s⁻¹).

3. Broadening function: LL Aqr and AK For

	LL Aqr		AK For	
	Primary	Secondary	Primary	Secondary
Nights compared	16	16	14	14
Offset Δ (km s ⁻¹)	$+0.251 \pm 0.023$	$+0.294 \pm 0.040$	$+0.208 \pm 0.059$	$+0.318 \pm 0.130$
Scatter s (km s ⁻¹)	0.092	0.142	0.150	0.864
Reduced χ^2	1.03	0.81	0.45	2.54
Slope a	0.9997	0.9989	0.9986	0.9923
Amplitude error	-0.03%	-0.11%	-0.14%	-0.77%
Authors' own fit rms	0.053	0.049	0.082	0.096

Table 2. Broadening-function velocities against the published ones. The last row is the residual rms of the authors' own orbital fits.

LL Aqr (16 nights). Offsets $+0.251 \pm 0.023$ and $+0.294 \pm 0.040$ km s⁻¹, scatter 0.092 and 0.142 km s⁻¹, χ_r^2 1.03 and 0.81, amplitude error -0.03% and -0.11% (primary, secondary).

AK For (14 nights: 12 FEROS, 2 HARPS). Offsets $+0.208 \pm 0.059$ and $+0.318 \pm 0.130$ km s⁻¹, scatter 0.150 and 0.864 km s⁻¹, χ_r^2 0.45 and 2.54, amplitude error -0.14% and -0.77% (primary, secondary). Secondary residuals range from -1.5 km s⁻¹ near phase 0.1 to $+1.3$ km s⁻¹ beyond phase 0.6.

4. Cross-correlation: 16 UVES high-velocity stars

Star	RV, this work [km s ⁻¹]	V _r meas [km s ⁻¹]	V _r Gaia DR3 [km s ⁻¹]	Δ [km s ⁻¹]
TYC 7524-00065-1	+351.498 ± 0.027	+351.920	+350.953 ± 0.217	-0.422
TYC 7535-00160-1	+355.074 ± 0.037	+353.523	+355.069 ± 0.253	+1.551
C0213360-505024	+387.999 ± 0.027	+388.600	+388.222 ± 0.611	-0.601
C1012254-203007	+508.139 ± 0.035	+506.755	+508.138 ± 1.068	+1.384
C1032508-244851	+509.755 ± 0.029	+509.490	+515.354 ± 2.470	+0.265
C1126364-293415	+411.154 ± 0.019	+409.870	+409.674 ± 0.926	+1.284
C1250398-030748	+398.245 ± 0.031	+403.020	+396.221 ± 0.979	-4.775
C1302091-323721	+410.550 ± 0.021	+410.000	+409.817 ± 3.027	+0.550
C1304064-331913	+360.461 ± 0.019	+360.720	+357.825 ± 0.994	-0.259
C1330590-230955	+335.542 ± 0.027	+335.350	+334.919 ± 0.526	+0.192
TYC 7274-00734-1	+408.425 ± 0.020	+408.317	+407.955 ± 0.208	+0.108
C1519196-191359	-409.410 ± 0.026	-409.610	-409.694 ± 6.570	+0.200
C1536201-144228	-329.600 ± 0.023	-329.600	-328.647 ± 2.112	-0.000
C1919566-632839	+330.810 ± 0.032	+328.680	+330.399 ± 0.530	+2.130
C2136142-694908	+383.373 ± 0.017	+381.640	+382.882 ± 0.330	+1.733
TYC 8019-00159-1	+99.097 ± 0.035	+99.117	+98.450 ± 0.171	-0.020

Table 3. Row for row in the order of Table 3 of Alazzawi et al. (2026). Our velocities are barycentric; ± is the formal error of the Gaussian fit to the CCF peak. V_r^{meas} is their template-matching velocity from the same spectrum (no per-star error published), and $\Delta = \text{ours} - V_r^{\text{meas}}$. $V_r^{\text{Gaia DR3}}$ is listed for reference only and enters no statistic.

	All 16 stars	Excluding C1250398-030748
Weighted offset Δ (km s ⁻¹)	+0.207 ± 0.125	+0.539 ± 0.129
Median Δ (km s ⁻¹)	0.486	0.422
Scatter s (km s ⁻¹)	1.564	0.854
Within 1 km s ⁻¹	10 of 16	10 of 15
Slope a	1.000171 ± 0.000479	1.000771 ± 0.000482
Velocity-scale error a − 1	+0.017 ± 0.048%	+0.077 ± 0.048%

Table 4. Summary of Table 3, with and without the largest difference.

All 16 stars. Weighted offset +0.207 ± 0.125 km s⁻¹, median |Δ| 0.486 km s⁻¹, scatter 1.564 km s⁻¹, 8 stars within 0.5 km s⁻¹ and 10 within 1 km s⁻¹, slope 1.000171 ± 0.000479 (+0.017 ± 0.048%) over 919 km s⁻¹.

Largest difference. C1250398-030748, Δ = -4.775 km s⁻¹ (next largest 2.130 km s⁻¹); S/N 161, CCF peak significance 31.8. Without it: offset +0.539 ± 0.129 km s⁻¹, scatter 0.854 km s⁻¹, slope 1.000771 ± 0.000482.

References

- Alazzawi, J. M. K., Caffau, E., Bonifacio, P., et al. 2026, A&A, DOI 10.1051/0004-6361/202659789
 Graczyk, D., Smolec, R., Gazeas, K., et al. 2016, A&A, 594, A92
 Hełminiak, K. G., Graczyk, D., Konacki, M., et al. 2014, A&A, 567, A64
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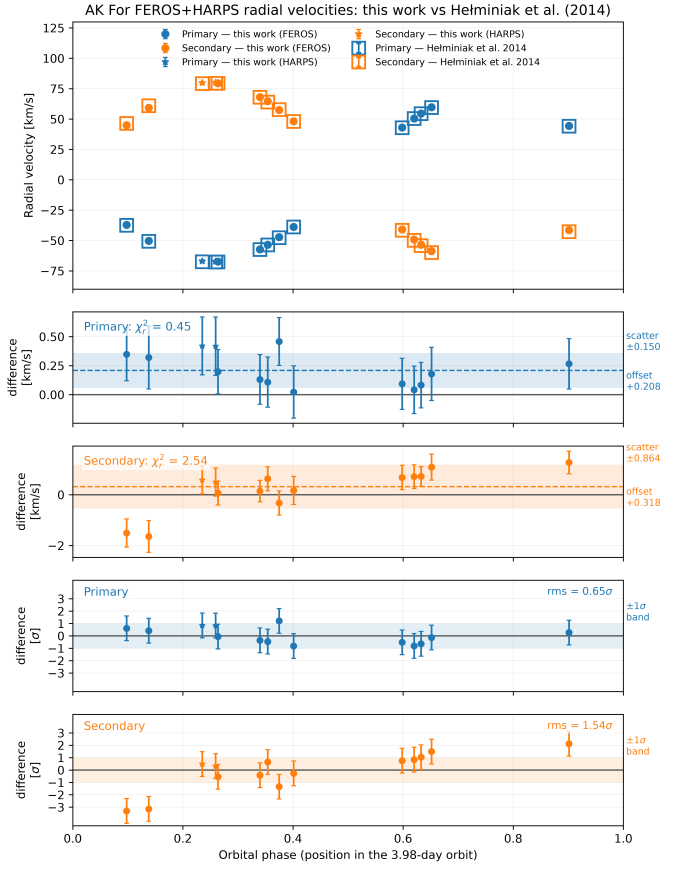
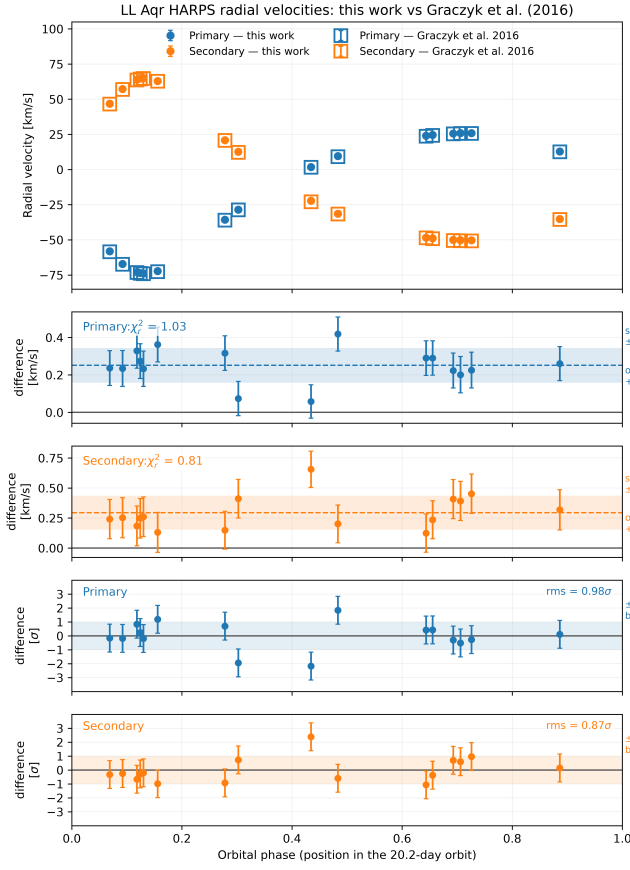


Figure 1. Broadening-function velocities against the published ones: LL Aqr versus Graczyk et al. 2016 (left, 16 nights) and AK For versus Helminiak et al. 2014 (right, 14 nights; circles FEROS, stars HARPS). Top: phased velocity curves (filled: this work, open: published). Middle: differences with the weighted offset (dashed) and scatter (band). Bottom: residuals in units of their own 1σ after removing the offset.